Problem 5: Thinning Property and Independence

Detailed Explanation

The Key Question: When Does Thinning Give Independence?

The **thinning property** is crucial for understanding Problem 5. Let's clarify when it applies.

Thinning Property

If $\{N_t\}$ is a Poisson process with rate λ, and each arrival is independently classified as "Type A" with probability p or "Type B" with probability $1 - p$, then:

- Type A arrivals form a Poisson process with rate λp
- Type B arrivals form a Poisson process with rate $\lambda(1 - p)$
- The two processes are **independent**

Problem 5a: Fixed n (NO Thinning, NOT Independent)

Setup:

- n button presses total (FIXED)
- Each press: Red with prob $1/3$, Green with prob $2/3$
- R_n = number of red presses, G_n = number of green presses

Why R_n and G_n are NOT Independent

Constraint: $R_n + G_n = n$ always holds

This is a **deterministic constraint**:

- If $R_n = 5$ and $n = 10$, then $G_n = 5$ with probability 1
- Knowing R_n tells you G_n exactly
- Perfect negative correlation

Mathematical proof of dependence:
For independence, we need:

$$P(R_n = r, G_n = g) = P(R_n = r) \cdot P(G_n = g) \quad \text{for all } r, g$$

But this **fails**. Consider $n = 10$, $r = 3$, $g = 4$:
Left side:

$$P(R_{10} = 3, G_{10} = 4) = 0 \quad (\text{since } 3 + 4 \neq 10)$$

Right side:

$$P(R_{10} = 3) = \binom{10}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^7 > 0$$
$$P(G_{10} = 4) = \binom{10}{4} \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^6 > 0$$
$$P(R_{10} = 3) \cdot P(G_{10} = 4) > 0$$

So $0 \neq P(R_{10} = 3) \cdot P(G_{10} = 4)$, violating independence.

Answer for 5a

Joint Distribution:

$$P(R_n = r, G_n = g) = \begin{cases} \binom{n}{r} \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{n-r} & \text{if } r + g = n \\ 0 & \text{otherwise} \end{cases}$$

Independence: NO, R_n and G_n are **NOT independent**

Why Doesn't Thinning Apply Here?

Thinning applies to **Poisson processes** (continuous time, random number of arrivals).
In Problem 5a:

- We have a **fixed** number n of trials
- This is **binomial**, not Poisson
- No thinning property

Think of it this way: With $n = 10$ button presses, once you count the red presses, the number of green presses is determined. There's no randomness left!

Problem 5b: Random N (Thinning DOES Apply, ARE Independent)

Setup:

- $N \sim \text{Poisson}(a)$ total presses (RANDOM)
- Each press: Red with prob 1/3, Green with prob 2/3
- R_N = number of red presses, G_N = number of green presses

Why R_N and G_N ARE Independent

Now we **can** apply the thinning property!

Think of this as a Poisson process:

- Start with a Poisson process of button presses with rate λ over time $[0, T]$ where $\lambda T = a$
- Each arrival is independently colored red (prob 1/3) or green (prob 2/3)
- By the **thinning property**:
  - Red presses form a Poisson process with rate $\lambda \cdot \frac{1}{3}$
  - Green presses form a Poisson process with rate $\lambda \cdot \frac{2}{3}$
  - These two processes are **independent**

Over the interval, the total counts are:

$$R_N \sim \text{Poisson}\left(\frac{a}{3}\right)$$

$$G_N \sim \text{Poisson}\left(\frac{2a}{3}\right)$$

And they are **independent**!

Answer for 5b

Joint Distribution:

$$P(R_N = r, G_N = g) = \frac{(a/3)^r e^{-a/3}}{r!} \cdot \frac{(2a/3)^g e^{-2a/3}}{g!}$$

Independence: YES, R_N and G_N **ARE independent**

Why Is This Different?

The key difference:

- In 5a: n is **fixed**, so $R_n + G_n = n$ is a rigid constraint
- In 5b: N is **random**, so $R_N + G_N = N$ is itself random
- The randomness of N breaks the dependency!

Intuition:

- With fixed n: Once you know how many reds, you know how many greens (no randomness left)
- With random N: Even if you know $R_N = 5$, you don't know G_N because you don't know N

Summary Table

Problem	Total	Thinning?	Independent?
5a: R_n, G_n	Fixed n	No (Binomial)	**NO**
5b: R_N, G_N	Random $N \sim \text{Poisson}(a)$	Yes (Poisson)	**YES**

Key Takeaway

Thinning property gives independence when:

- You start with a **Poisson process**
- You randomly classify arrivals

It does **NOT** apply when:

- You have a **fixed** number of trials (binomial setting)
- The constraint $R_n + G_n = n$ makes them dependent